

INHA UNIVERSITY IN TASHKENT

School of Computers and Information Engineering

Computer Algorithms (CIE3090)

Fall semester 2024

Submission Deadline: 23 /09/ 2024 (23:30 Hr) - Monday

ID _____ Name _____ Gp_____

NOTE:

1. You are required to **take print out of this page** and solve (**handwritten**) the questions on this sheet only. [Submissions not conforming to the given format will not be evaluated]
2. **Copying** of solution is strictly prohibited. In case it is found, then zero marks will be awarded.
3. Submit scan copy on eclass. Hardcopy submission not required. Late submissions are not accepted.

Questions

Q1. Find the O- notation for following functions. (Show the steps to evaluate value of c and no)

a. $f(n) = 6n^3 + 6n + 2$

b. $f(n) = n + 2^n$

Q2. Find the big omega- notation for following functions. (Show the steps)

a. $n^3 + n^2 + 2$

b. $7 \cdot 2^n + 6n^3$

Q3. Find the theta notation for following functions. (Show the steps)

a. $f(n) = 2^n + 4n^3 + n$

Q4 Comment on the following (check bounds are correct or incorrect)

Given $f(n) = 5n^3 + 2$ and $g(n) = n^3$

a. $g(n) = o(f(n))$ [o is small oh]

b. $f(n) = \Omega(g(n))$ [Ω is big omega]